

A Squared-Complex Framework for Square-Prefix Möbius Sums

Exact arithmetic, canonical cofactor geometry, the elementary pointwise bridge,
and axiom-free Lean formalization

Fred Viole
OVVO Labs

July 2026

Abstract

This paper develops an exact geometric framework for square-prefix values of the Mertens function. At the complete cutoff

$$X_n = (n + 1)^2 - 1,$$

a squarefree integer cannot contain two prime factors larger than $n + 1$. This gives the exact smooth-transport decomposition

$$S_n = M(X_n) = A_n - T_n,$$

where A_n is the signed mass already smooth at the square-root cutoff and T_n records sources with one prime factor still beyond it.

Each squarefree source $m > 1$ has a unique canonical factor pair

$$q_m = P^+(m), \quad c_m = \frac{m}{q_m}.$$

Writing

$$a_m = \frac{c_m + q_m}{2}, \quad b_m = \frac{q_m - c_m}{2},$$

gives

$$(a_m + b_m i)^2 = m + iY_m, \quad Y_m = \frac{q_m^2 - c_m^2}{2}.$$

The real coordinate is the source integer and the imaginary coordinate records canonical factor imbalance. Fixed cofactors trace explicit parabolas, the prefix cutoff is a moving vertical boundary, and the smoothness cutoff is a moving parabola. Their intersection occurs exactly at the arithmetic transition $q = R$, so internal transport-to-smooth transfers telescope identically in $A - T$.

For a fixed height threshold $\Lambda > 0$, the canonical source population splits into low and high sectors according to $|Y_m| \leq \Lambda e(m)$ or $|Y_m| > \Lambda e(m)$, where $e(m) = \lfloor \sqrt{m} \rfloor$. Low height forces a small factor gap, bounded square-block occupancy, and bounded transport lifetime. This yields the elementary pointwise bound

$$|S_N^{\text{low}}(\Lambda)| \ll_{\Lambda} N.$$

The latest reduction removes every unnecessary averaging parameter from the remaining logical burden. For each fixed $\Lambda > 0$, the Riemann Hypothesis is equivalent to the single finite-sum inequality

$$\boxed{\left| S_N^{\text{high}}(\Lambda) \right|^2 \ll_{\varepsilon, \Lambda} N^{2+\varepsilon}.$$

Equivalently, because $S_N = A_N - T_N$ and the low sector is elementary,

$$|A_N - T_N| \ll_\varepsilon N^{1+\varepsilon}.$$

The formerly emphasized translated-window estimate follows by summing the pointwise bound over $N \leq n < N + H \leq 2N$, and conversely gives the pointwise bound by taking $H = 1$.

The accompanying Lean development formalizes the exact arithmetic and geometry, the signed Hilbert identities, the concrete square-prefix Mertens bridge, and the native largest-prime-factor low/high realization. The hard analytic bound and the classical Mertens–RH theorem remain explicit ordinary typed premises. No project axiom and no unconditional proof of RH is claimed.

How to read this paper

The paper has three layers. First, the square-root cutoff gives an exact arithmetic decomposition. Second, the unique largest-prime-factor pair is embedded in squared complex coordinates, producing explicit cofactor curves, entry times, transition times, and a canonical height. Third, the low sector is closed by an elementary occupancy estimate, reducing the entire remaining theorem to one pointwise high-sector inequality. The Lean development records exactly which of these steps are machine checked.

Readers interested mainly in the RH connection may begin with [Sections 1, 14, 15, 17, 19 and 20](#). Readers interested in the geometry may begin with [Sections 3, 4, 6 and 7](#). The numerical section is evidence and diagnosis, not a substitute for the missing pointwise theorem.

1 Overview and main results

Why this section is here. The choice of square prefixes is not cosmetic. It gives a cutoff with an exact square root, which turns a complicated smoothness condition into a clean one-large-prime decomposition. The squared complex coordinate is then chosen because it records the product and factor imbalance at the same time.

Set

$$X_n = (n + 1)^2 - 1, \quad S_n = M(X_n).$$

Why stop immediately before a square? Because if $m < R^2$ is squarefree, it cannot contain two prime factors larger than R . Every source is therefore either already R -smooth or has a unique representation

$$m = cq, \quad c < R < q,$$

with q prime. This gives the exact identity

$$S_n = A_n - T_n.$$

The identity is elementary; the difficult question is why the two much larger terms A_n and T_n track each other so closely.

The geometry is designed to make that question visible. For the factor pair $cq = m$, define

$$a = \frac{c + q}{2}, \quad b = \frac{q - c}{2}.$$

Then

$$(a + bi)^2 = m + i \frac{q^2 - c^2}{2}.$$

The real coordinate remembers the integer m , while the imaginary coordinate measures how uneven the factor pair is. Fixed cofactors become parabolas. The complete prefix is cut out by a vertical line. Smoothness is cut out by a second moving parabola. A source first enters the prefix, then spends time in the transport region, and finally becomes smooth when the moving boundary reaches its largest prime factor.

This viewpoint separates three logically different mechanisms:

1. *exact bookkeeping*: the transport-to-smooth crossing cancels identically in $A - T$;
2. *geometric dilution*: the quadratic map expands area by order n^2 at square scale n ;
3. *analytic cancellation*: the remaining high-height contribution must be controlled through one signed Gram form, including its off-diagonal terms.

The first two are proved here. The third is the central open estimate. The coordinate change does not prove cancellation by itself; it identifies the correct state space, scale, and interaction that a proof must control.

The main exact results are summarized below.

Theorem 1.1 (Exact square-root decomposition). *For $R_n = n + 1$ and $X_n = R_n^2 - 1$, define*

$$A_n := \sum_{\substack{m \leq X_n \\ P^+(m) \leq R_n}} \mu(m),$$

$$T_n := \sum_{c < R_n} \mu(c) \left[\pi \left(\left\lfloor \frac{X_n}{c} \right\rfloor \right) - \pi(R_n) \right].$$

Then

$$\boxed{S_n = A_n - T_n.}$$

Theorem 1.2 (Squared-space recovery). *Let $w = x + iy$ be generated by a positive factor pair $cq = x$ through*

$$y = \frac{q^2 - c^2}{2}.$$

With $\rho = |w|$,

$$\boxed{c^2 = \rho - y, \quad q^2 = \rho + y.}$$

Hence the squared point determines the ordered factor pair exactly.

Theorem 1.3 (Cofactor-curve cutoff intersection). *For fixed $c > 0$, put*

$$\gamma_c(x) = \frac{x^2 - c^4}{2c^2}.$$

At cutoff $R > 0$, put

$$\Gamma_R(x) = \frac{R^4 - x^2}{2R^2}.$$

Then

$$\boxed{\gamma_c(x) = \Gamma_R(x) \iff x = cR.}$$

The curves meet orthogonally at their intersection. At an arithmetic point $x = cq$, the sign of $\gamma_c(cq) - \Gamma_R(cq)$ is the sign of $q - R$.

Theorem 1.4 (Exact channel telescoping). *As R increases, a canonical point enters the prefix at $R = \sqrt{cq}$ and crosses from transport to smooth at $R = q$. The signed mass crossing the moving parabola appears with opposite signs in the increments of A and $-T$. Therefore all internal channel transfers cancel from the increment of $A - T$; only points entering the new square strip remain.*

Theorem 1.5 (Low-height clustering and lifetime). *Let $m = cq \in B_n$ be canonical and let*

$$Y_m = \frac{q^2 - c^2}{2}.$$

Then, for $H \geq 0$,

$$\#\{m \in B_n : |Y_m| \leq H\} \leq \left\lfloor \frac{H}{n} \right\rfloor,$$

and the count is zero when $H = n$. For odd sources the bound improves to $\lfloor H/(2n) \rfloor$. If $Y_m > 0$, the transport lifetime satisfies

$$L(m) \leq \frac{2Y_m}{q}.$$

Consequently, for fixed Λ , the canonical points with $|Y_m| \leq \Lambda n$ have uniformly bounded block occupancy, and the transport members have uniformly bounded lifetime.

Theorem 1.6 (Elementary canonical pointwise criterion). *Fix $\Lambda > 0$. Extend the harmless conventions $P^+(1) = 1$, $c_1 = 1$, and $Y_1 = 0$. For the square block $B_j = [j^2, (j+1)^2)$ define*

$$d_j^{\text{high}}(\Lambda) := \sum_{\substack{m \in B_j \\ |Y_m| > \Lambda j}} \mu(m), \quad S_N^{\text{high}}(\Lambda) := \sum_{0 \leq j \leq N} d_j^{\text{high}}(\Lambda).$$

Then the following are equivalent:

(i) *the Riemann Hypothesis;*

(ii) *for every $\varepsilon > 0$,*

$$\boxed{|S_N^{\text{high}}(\Lambda)|^2 \ll_{\varepsilon, \Lambda} N^{2+\varepsilon};} \tag{PHS}$$

(iii) *for every $\varepsilon > 0$ and $1 \leq H \leq N$,*

$$\sum_{n=N}^{N+H-1} |S_n^{\text{high}}(\Lambda)|^2 \ll_{\varepsilon, \Lambda} H N^{2+\varepsilon}. \tag{HS}$$

The equivalence uses only the exact low/high recombination, the elementary low-sector bound, square-endpoint interpolation, and the classical Mertens criterion. The implication (ii) $\Rightarrow$ (iii) follows from $N + h < 2N$; the reverse implication is the single choice $H = 1$.

Theorem 1.7 (Quadratic conformal expansion). *The map*

$$\Phi(c, q) = \left(cq, \frac{q^2 - c^2}{2} \right)$$

has

$$D\Phi(c, q)^T D\Phi(c, q) = (c^2 + q^2)I,$$

and

$$\det D\Phi(c, q) = c^2 + q^2 \geq 2cq.$$

At square scale $cq \asymp n^2$, lengths expand by at least order n and areas by at least order n^2 .

Theorem 1.8 (Exact dyadic cell compression). *For every $k \geq 0$,*

$$\boxed{(\mu(4k+1), \mu(4k+2), \mu(4k+3), \mu(4k+4)) = (\mu(4k+1), -\mu(2k+1), \mu(4k+3), 0)}.$$

Consequently the complete-cell scalar is

$$\boxed{C_k = \mu(4k+1) - \mu(2k+1) + \mu(4k+3)}.$$

Among the three active slots $4k+1, 4k+2, 4k+3$, exactly one is divisible by 3; the affected slot is $4k+3, 4k+2, 4k+1$ according as $k \equiv 0, 1, 2 \pmod{3}$.

Theorem 1.9 (Exact small-modulus quadratic resonance). *For every prime $q > 3$,*

$$\boxed{q^2 \equiv 1 \pmod{24}}.$$

Hence, for every integer m and every prime interval containing no 2 or 3,

$$\boxed{\sum_{\substack{Q < q \leq 2Q \\ q \text{ prime}}} e\left(\frac{mq^2}{24}\right) = e\left(\frac{m}{24}\right) (\pi(2Q) - \pi(Q))}.$$

Equivalently, the quadratic prime phase is perfectly coherent at every $\alpha = m/12$. More generally, for $\alpha = a/r$, the rational phase $e(aq^2/(2r))$ is periodic modulo $2r$; the correct reduced-residue model is therefore formed over $(\mathbb{Z}/2r\mathbb{Z})^\times$.

The remaining theorem is the pointwise inequality (PHS). The local estimate (HS) is an exact equivalent. No claim is made that the coordinate change alone supplies cancellation; it identifies the native arithmetic quantity, the correct scale, and the signed interaction that must be controlled.

2 Square prefixes and the arithmetic decomposition

Why this section is here. The square-root cutoff is the arithmetic hinge of the paper. Before introducing geometry, we isolate the exact decomposition that the geometry is meant to explain.

Define the complete square block

$$B_n := \{m \in \mathbb{N} : n^2 \leq m < (n+1)^2\}, \tag{1}$$

the square-root cutoff

$$R_n := n + 1, \quad (2)$$

and the complete-prefix endpoint

$$X_n := R_n^2 - 1 = (n + 1)^2 - 1. \quad (3)$$

The square-prefix Mertens value is

$$S_n := M(X_n) = \sum_{m \leq X_n} \mu(m). \quad (4)$$

Because $\mu(R_n^2) = 0$ for $R_n > 1$, one may equally write $S_n = M(R_n^2)$.

Write $P^+(m)$ for the largest prime factor of m , with $P^+(1) = 1$.

Definition 2.1 (Smooth and transport terms). Define

$$A_n := \sum_{\substack{1 \leq m \leq X_n \\ P^+(m) \leq R_n}} \mu(m), \quad (5)$$

$$T_n := \sum_{1 \leq c < R_n} \mu(c) \left[\pi\left(\left\lfloor \frac{X_n}{c} \right\rfloor\right) - \pi(R_n) \right]. \quad (6)$$

Lemma 2.2 (One-large-prime closure). *Let $m \leq R^2 - 1$ be squarefree. If $P^+(m) > R$, then there is a unique factorization*

$$m = cq, \quad 1 \leq c < R < q, \quad q \text{ prime}, \quad (7)$$

where $q = P^+(m)$. Moreover,

$$\mu(m) = -\mu(c). \quad (8)$$

Proof. Put $q = P^+(m)$ and $c = m/q$. If $c \geq R$, then $m = cq > R^2$, contrary to $m \leq R^2 - 1$. Since m is squarefree, $q \nmid c$ and $\mu(cq) = -\mu(c)$. Uniqueness follows from the largest-prime choice. $\square$

Proof of Theorem 1.1. Partition the squarefree $m \leq X_n$ according to whether $P^+(m) \leq R_n$ or $P^+(m) > R_n$. The first population contributes A_n . By Lemma 2.2, the second population is uniquely $m = cq$ with $c < R_n < q$, and each source has weight $-\mu(c)$. The total second contribution is therefore $-T_n$. $\square$

Corollary 2.3 (Elementary smooth-transport RH inequality). *The Riemann Hypothesis is equivalent to either of the following square-prefix estimates:*

$$|A_N - T_N|^2 \ll_\varepsilon N^{2+\varepsilon}, \quad (9)$$

$$|A_N - T_N| \ll_\varepsilon N^{1+\varepsilon}. \quad (10)$$

Here and below the two exponent conventions are equivalent after renaming ε .

Proof. By Theorem 1.1, $A_N - T_N = M((N + 1)^2 - 1)$. The classical Mertens criterion gives the estimates under RH. Conversely, every x lies within one square block of an endpoint $(N + 1)^2 - 1$, and $|\mu(m)| \leq 1$, so the endpoint estimate interpolates to $M(x) \ll_\varepsilon x^{1/2+\varepsilon}$. $\square$

Remark 2.4 (Why this is the most elementary total formulation). The inequality Equation (10) contains no shell, window, covariance, probability space, or operator. It asks only that the two exact cumulative arithmetic terms in $S_N = A_N - T_N$ agree to square-root scale. The high-sector formulation introduced later removes the part of this agreement already forced by elementary geometry.

2.1 The largest-prime cofactor identity

For $c \geq 1$, define the prime-tail counting function

$$\Pi_c(x) := \# \left\{ q \text{ prime} : P^+(c) < q \leq \left\lfloor \frac{x}{c} \right\rfloor \right\}. \quad (11)$$

Equivalently,

$$\Pi_c(x) = \left(\pi \left(\left\lfloor \frac{x}{c} \right\rfloor \right) - \pi(P^+(c)) \right)_+, \quad (12)$$

where $(u)_+ := \max\{u, 0\}$. The positive part is essential when $x/c < P^+(c)$.

Theorem 2.5 (Exact largest-prime cofactor expansion). *For every real $x \geq 1$,*

$$M(x) = 1 - \sum_{c \geq 1} \mu(c) \Pi_c(x). \quad (13)$$

The sum is finite, and separating the prime curve $c = 1$ gives

$$M(x) = 1 - \pi(x) - \sum_{2 \leq c \leq x/2} \mu(c) \left(\pi \left(\left\lfloor \frac{x}{c} \right\rfloor \right) - \pi(P^+(c)) \right)_+. \quad (14)$$

Proof. Every squarefree $m > 1$ has a unique largest prime factor $q = P^+(m)$. Writing $c = m/q$, one has

$$q > P^+(c), \quad \mu(m) = \mu(cq) = -\mu(c).$$

Conversely, every pair consisting of a squarefree c and a prime $q > P^+(c)$ produces the squarefree integer $m = cq$ whose largest prime factor is q . The condition $m \leq x$ is exactly $q \leq x/c$. Summing $-\mu(c)$ over these unique pairs proves Equation (13). The $c = 1$ term is $-\pi(x)$, because $P^+(1) = 1$. No term with $c > x/2$ can occur. $\square$

Remark 2.6 (Central analytic engine). Equation (14) is the exact arithmetic form of the cofactor-parabola decomposition. It shows that the Mertens function is not obtained by proving that the prime curve is small. The prime curve $-\pi(x)$ is cancelled by the Möbius-weighted collection of all curves $c \geq 2$.

3 All factor pairs above one integer

Why this section is here. A fixed integer has many factor pairs. This section describes the complete geometric fibre first, so that the later canonical Möbius point is not confused with the whole factorization geometry.

The factorization geometry and the smooth-transport geometry use the same coordinate map but different point selections. This distinction is essential.

Fix $N \geq 1$ and consider every positive factor pair

$$uv = N, \quad 1 \leq u \leq v. \quad (15)$$

Define the Fermat coordinates

$$a = \frac{u+v}{2}, \quad b = \frac{v-u}{2} \geq 0. \quad (16)$$

Then

$$u = a - b, \quad v = a + b, \quad N = a^2 - b^2. \quad (17)$$

The coordinates lie in $\frac{1}{2}\mathbb{Z}$; they are integers when u and v have the same parity.

Squaring the complex factor gives

$$(a + bi)^2 = N + iY, \quad \boxed{Y = 2ab = \frac{v^2 - u^2}{2}}. \quad (18)$$

Thus all factorizations of a fixed N lie on the vertical fibre

$$\operatorname{Re} w = N. \quad (19)$$

3.1 Square-root tests

Let

$$w = N + iY, \quad \rho = |w| = \sqrt{N^2 + Y^2}. \quad (20)$$

Then

$$a^2 = \frac{\rho + N}{2}, \quad (21)$$

$$b^2 = \frac{\rho - N}{2}, \quad (22)$$

$$u^2 = \rho - Y, \quad (23)$$

$$v^2 = \rho + Y. \quad (24)$$

Accordingly, a candidate vertical ordinate Y corresponds to an exact factorization if and only if the relevant square and parity conditions hold. The identity

$$\sqrt{N^2 + Y^2} = a^2 + b^2 \quad (25)$$

is the right-triangle form of the same condition.

3.2 Admissible intervals

For a factor pair $u(N/u) = N$ with $1 \leq u \leq \sqrt{N}$,

$$a(u) = \frac{u + N/u}{2}, \quad b(u) = \frac{N/u - u}{2}. \quad (26)$$

Both functions decrease as u increases toward $\sqrt{N}$. Therefore the full positive factor fibre satisfies

$$\sqrt{N} \leq a \leq \frac{N+1}{2}, \quad (27)$$

$$0 \leq b \leq \frac{N-1}{2}, \quad (28)$$

$$0 \leq Y \leq \frac{N^2 - 1}{2}. \quad (29)$$

For integer enumeration one uses $a \geq \lceil \sqrt{N} \rceil$ with the parity required by N .

The upper endpoint corresponds to the trivial factor pair $1 \cdot N$. If the trivial factor is excluded and the smallest permitted odd factor is 3, then

$$a \leq \frac{N + 9}{6}, \quad (30)$$

$$b \leq \frac{N - 9}{6}, \quad (31)$$

$$Y \leq \frac{N^2 - 81}{18}. \quad (32)$$

These are the nontrivial odd-factor or semiprime-search envelopes. They are not the universal bounds for the complete factor fibre.

Example 3.1 (The fibre $N = 309$). The trivial factor pair $1 \cdot 309$ gives

$$(a, b) = (155, 154), \quad w = 309 + 47740i.$$

The nontrivial factorization $3 \cdot 103$ gives

$$(a, b) = (53, 50), \quad w = 309 + 5300i.$$

The latter is the highest point allowed by the nontrivial odd-factor condition $u \geq 3$, but not the highest point on the full factor fibre.

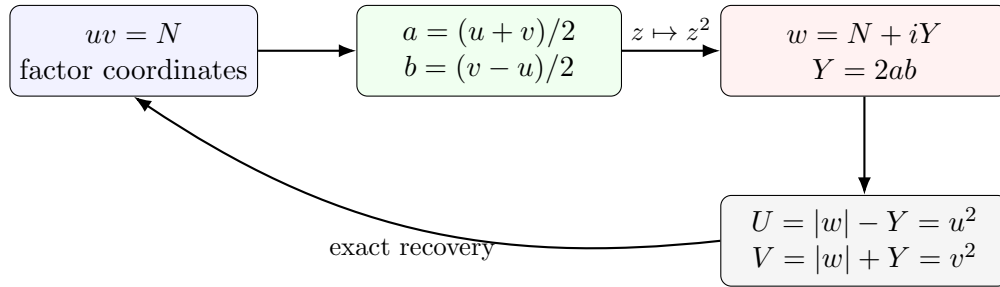


Figure 1: Equivalent coordinate systems on a complete factor fibre. The squared coordinates retain the factor pair exactly.

4 The canonical Möbius geometry

Why this section is here. The Möbius decomposition needs one reproducible point per squarefree source. Choosing the largest prime factor supplies that point and gives a signed height that can be followed across cutoffs.

The smooth-transport decomposition does not use every point on each factor fibre. It selects one canonical point per squarefree source.

For squarefree $m > 1$, define

$$q_m := P^+(m), \quad c_m := \frac{m}{q_m}. \quad (33)$$

Unlike the ordered factor pair in [Section 3](#), the canonical pair is not reordered. The cofactor c_m may exceed q_m . For example,

$$m = 30, \quad q_m = 5, \quad c_m = 6.$$

Therefore define the signed Fermat coordinate

$$a_m := \frac{c_m + q_m}{2}, \quad b_m := \frac{q_m - c_m}{2} \in \frac{1}{2}\mathbb{Z}, \quad (34)$$

and the canonical squared point

$$w_m := (a_m + ib_m)^2 = m + iY_m, \quad (35)$$

where

$$Y_m = 2a_mb_m = \frac{q_m^2 - c_m^2}{2}. \quad (36)$$

The sign of Y_m records the order of the canonical factors:

$$Y_m > 0 \iff c_m < q_m, \quad Y_m < 0 \iff c_m > q_m. \quad (37)$$

4.1 Parameterization by cofactors and primes

Every squarefree $m > 1$ has a unique representation

$$m = cq, \quad (38)$$

where

$$\mu(c) \neq 0, \quad q \text{ is prime}, \quad q > P^+(c). \quad (39)$$

Conversely, every pair satisfying [Equation \(39\)](#) produces a squarefree integer $m = cq$ with largest prime factor q . Moreover,

$$\mu(cq) = -\mu(c). \quad (40)$$

Define the canonical index set

$$\mathcal{C} := \{(c, q) : \mu(c) \neq 0, \ q \text{ prime}, \ q > P^+(c)\}. \quad (41)$$

The squared point cloud is

$$\mathcal{W} := \left\{ cq + i \frac{q^2 - c^2}{2} : (c, q) \in \mathcal{C} \right\}. \quad (42)$$

The signed point measure is

$$\nu := \delta_1 - \sum_{(c,q) \in \mathcal{C}} \mu(c) \delta_{cq + i(q^2 - c^2)/2}. \quad (43)$$

The atom at 1 records $\mu(1) = 1$.

Example 4.1 (A point below the real axis). For $m = 30$, the canonical pair is $(c, q) = (6, 5)$, so

$$a = \frac{11}{2}, \quad b = -\frac{1}{2}, \quad w_{30} = 30 - \frac{11}{2}i.$$

This point enters a square prefix only after the cutoff already exceeds $q = 5$, so it is smooth immediately and never occupies the transport channel.

5 Four equivalent coordinate systems

Why this section is here. Different calculations become simple in different coordinates. The point is not to multiply notation, but to move between product, factor-gap, norm, and null coordinates without losing information.

Let

$$w = x + iy = cq + i\frac{q^2 - c^2}{2}, \quad \rho := |w| = \sqrt{x^2 + y^2}. \quad (44)$$

Then

$$\rho = \frac{c^2 + q^2}{2}. \quad (45)$$

The most useful coordinate charts are collected in [Table 1](#).

Table 1: Equivalent coordinates for a canonical squared factor point.

Coordinate system	Definition	Exact inverse information
Factor coordinates	(c, q)	$m = cq, q = P^+(m)$
Fermat coordinates	$a = (c + q)/2, b = (q - c)/2$	$c = a - b, q = a + b$
Squared Cartesian	$x = cq, y = (q^2 - c^2)/2$	$w = x + iy = (a + ib)^2$
Radial-null	$U = \rho - y, V = \rho + y$	$U = c^2, V = q^2, x^2 = UV$
Square-root	$a^2 = (\rho + x)/2, b^2 = (\rho - x)/2$	sign of b is sign of y

The central identities are

$$\boxed{U := \rho - y = c^2, \quad V := \rho + y = q^2.} \quad (46)$$

Consequently,

$$x = \sqrt{UV} = cq, \quad y = \frac{V - U}{2}. \quad (47)$$

These null coordinates linearize the arithmetic cutoffs. They also make the meaning of $2ab$ exact:

$$\boxed{Y = 2ab = \frac{V - U}{2},} \quad (48)$$

so the imaginary coordinate is half the separation between the squared canonical factors.

6 Fixed-cofactor parabolas

Why this section is here. Once the cofactor is fixed, the canonical points are no longer an unstructured cloud. They lie on explicit curves, which lets arithmetic transitions be read geometrically.

Fix a positive cofactor c and let q vary. In squared Cartesian coordinates,

$$x = cq, \quad y = \frac{q^2 - c^2}{2}. \quad (49)$$

Eliminating $q = x/c$ gives the cofactor parabola

$$\boxed{\gamma_c(x) = \frac{x^2 - c^4}{2c^2}}. \quad (50)$$

The canonical point cloud is not the filled region between envelopes. It is the arithmetic sampling

$$x = cq, \quad q \text{ prime}, \quad q > P^+(c), \quad (51)$$

on the family [Equation \(50\)](#).

The curve crosses the real axis at

$$x = c^2. \quad (52)$$

Points to the right have $q > c$ and positive ordinate; points to the left have $q < c$ and negative ordinate.

The $c = 1$ curve is

$$\gamma_1(x) = \frac{x^2 - 1}{2}. \quad (53)$$

Its arithmetic samples are exactly the prime points, since $m = 1 \cdot q = q$. It is the outer positive envelope of the complete canonical cloud. The curve

$$\gamma_3(x) = \frac{x^2 - 81}{18} \quad (54)$$

is the first nontrivial odd-cofactor envelope used in semiprime search.

7 Two moving boundaries

Why this section is here. The decomposition changes with the cutoff. The goal here is to represent that changing arithmetic classification as motion of two simple boundaries through one fixed point cloud.

Fix $R > 0$. The complete arithmetic prefix is

$$\mathcal{P}_R := \{w = x + iy \in \mathcal{W} : x < R^2\}. \quad (55)$$

Since $V = q^2 = |w| + y$, the smooth condition $q \leq R$ is

$$|w| + y \leq R^2. \quad (56)$$

Define

$$\mathcal{A}_R := \{w \in \mathcal{P}_R : |w| + \operatorname{Im} w \leq R^2\}, \quad (57)$$

$$\mathcal{U}_R := \{w \in \mathcal{P}_R : |w| + \operatorname{Im} w > R^2\}. \quad (58)$$

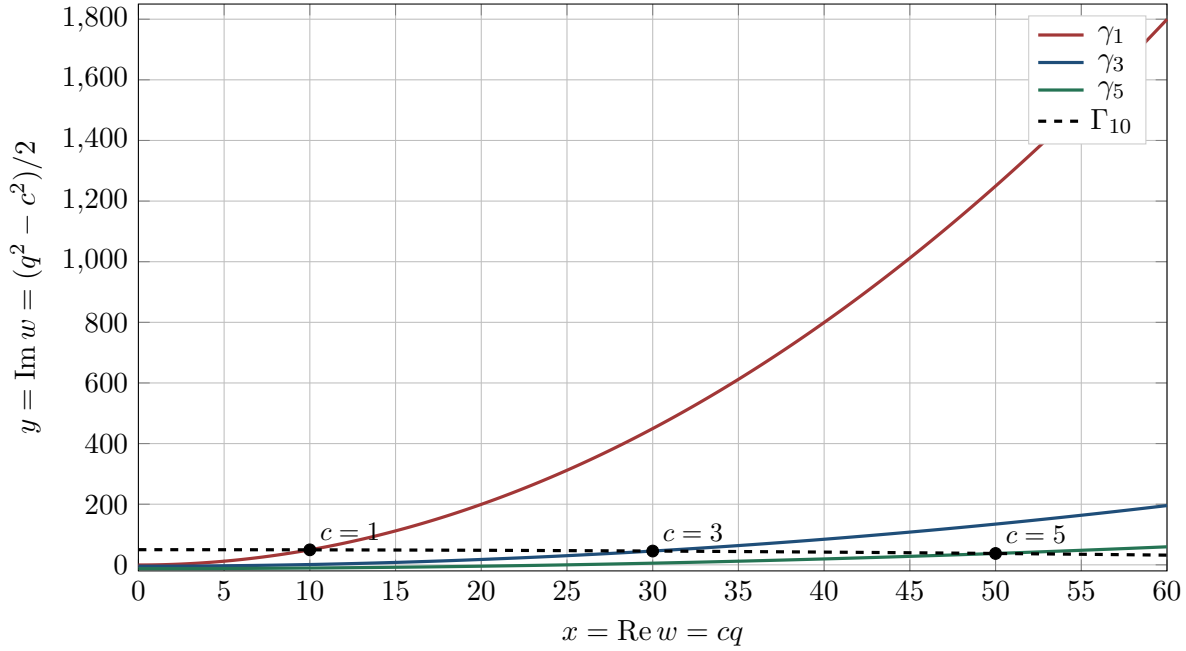


Figure 2: Cofactor parabolas γ_c and the moving cutoff parabola Γ_R for $R = 10$. Each γ_c meets Γ_R at $x = cR$. Actual canonical points are discrete prime samples on the curves.

The prefix is the disjoint union

$$\mathcal{P}_R = \mathcal{A}_R \dot{\cup} \mathcal{U}_R. \quad (59)$$

Solving Equation (56) on the boundary gives

$$\Gamma_R(x) = \frac{R^4 - x^2}{2R^2}. \quad (60)$$

Inside the prefix strip $x < R^2$, the smooth region lies below the graph and the transport region lies above it.

In null coordinates, the same regions are even simpler:

$$\mathcal{P}_R: \quad UV < R^4, \quad (61)$$

$$\mathcal{A}_R: \quad UV < R^4, \quad V \leq R^2, \quad (62)$$

$$\mathcal{U}_R: \quad UV < R^4, \quad V > R^2. \quad (63)$$

For a transport point, $UV < R^4 < V^2$, so automatically $U < R^2$. Thus

$$\mathcal{U}_R: \quad U < R^2 < V, \quad UV < R^4. \quad (64)$$

This is exactly $c < R < q$.

7.1 Intersection with every cofactor curve

Proposition 7.1 (Intersection factorization). *For $c, R > 0$,*

$$\gamma_c(x) - \Gamma_R(x) = \frac{(R^2 + c^2)(x^2 - c^2 R^2)}{2c^2 R^2}. \quad (65)$$

Consequently,

$$\gamma_c(x) = \Gamma_R(x) \iff x = cR. \quad (66)$$

At the arithmetic point $x = cq$,

$$\gamma_c(cq) - \Gamma_R(cq) = \frac{(R^2 + c^2)(q^2 - R^2)}{2R^2}. \quad (67)$$

The sign is therefore exact:

$$q < R \iff w_{c,q} \text{ lies below } \Gamma_R, \quad (68)$$

$$q > R \iff w_{c,q} \text{ lies above } \Gamma_R. \quad (69)$$

Proposition 7.2 (Orthogonal crossing). *At the intersection $x = cR$,*

$$\gamma'_c(cR) = \frac{R}{c}, \quad \Gamma'_R(cR) = -\frac{c}{R}, \quad (70)$$

so

$$\gamma'_c(cR)\Gamma'_R(cR) = -1. \quad (71)$$

Hence each cofactor curve meets the moving cutoff orthogonally.

The transition is therefore transverse, never tangent. A canonical point cannot remain trapped in a thin boundary layer as the cutoff passes its largest prime.

8 The moving-region interpretation of $A - T$

Why this section is here. This section translates the algebraic identity $A - T$ into a geometric partition. It is the bridge between the original Mertens sum and the later kernel formulation.

Using the signed measure [Equation \(43\)](#), define

$$\mathcal{A}(R) := \nu(\mathcal{A}_R), \quad \mathcal{U}(R) := \nu(\mathcal{U}_R). \quad (72)$$

At $R = R_n$,

$$\mathcal{A}(R_n) = A_n, \quad \mathcal{U}(R_n) = -T_n. \quad (73)$$

Indeed, the transport points satisfy $c < R_n < q$ and carry signed point mass

$$\mu(cq) = -\mu(c).$$

Therefore

$$\boxed{S_n = \nu(\mathcal{P}_{R_n}) = \mathcal{A}(R_n) + \mathcal{U}(R_n) = A_n - T_n.} \quad (74)$$

Equivalently, define projectors

$$P_R(w) := \mathbf{1}_{\{\operatorname{Re} w < R^2\}}, \quad (75)$$

$$Q_R(w) := \mathbf{1}_{\{|w| + \operatorname{Im} w \leq R^2\}}. \quad (76)$$

Then

$$\mathcal{A}(R) = \int P_R Q_R d\nu, \quad (77)$$

$$\mathcal{U}(R) = \int P_R (1 - Q_R) d\nu, \quad (78)$$

$$S(R) = \int P_R d\nu. \quad (79)$$

The identity is the pointwise partition

$$P_R = P_R Q_R + P_R (1 - Q_R). \quad (80)$$

9 Entry, transition, and telescoping

Why this section is here. A source contributes at different times for different reasons. Separating entry from transition shows which changes are genuine new data and which are internal transfers that cancel.

For a canonical source $m = cq$, define its first square-prefix index

$$e(m) := \lfloor \sqrt{m} \rfloor. \quad (81)$$

This is the least n such that $m < R_n^2$. Define the smoothness-transition index

$$h(m) := q - 1. \quad (82)$$

This is the least n such that $q \leq R_n$.

The transport lifetime is

$$L(m) := (h(m) - e(m))_+. \quad (83)$$

There are two cases:

1. If $q > \sqrt{m}$, equivalently $q > c$ and $Y > 0$, then the point enters above the moving parabola and occupies transport for the discrete interval

$$e(m) \leq n < h(m).$$

2. If $q \leq \sqrt{m}$, equivalently $q \leq c$ and $Y \leq 0$, then $h(m) \leq e(m)$ and the point enters already smooth. Its transport interval is empty.

Define the pointwise indicators

$$p_n(m) := \mathbf{1}_{\{e(m) \leq n\}}, \quad (84)$$

$$u_n(m) := \mathbf{1}_{\{e(m) \leq n < h(m)\}}, \quad (85)$$

$$a_n(m) := \mathbf{1}_{\{e(m) \leq n, h(m) \leq n\}}. \quad (86)$$

Then

$$\boxed{p_n(m) = u_n(m) + a_n(m)}. \quad (87)$$

9.1 Why the crossing cancels

Let E_n^A be the signed mass entering the new vertical strip already below the moving parabola, let E_n^U be the signed mass entering above it, and let C_n be the signed mass crossed by the moving parabola between R_{n-1} and R_n . Then

$$\mathcal{A}(R_n) - \mathcal{A}(R_{n-1}) = E_n^A + C_n, \quad (88)$$

$$\mathcal{U}(R_n) - \mathcal{U}(R_{n-1}) = E_n^U - C_n. \quad (89)$$

Therefore

$$S_n - S_{n-1} = E_n^A + E_n^U = \sum_{m \in B_n} \mu(m). \quad (90)$$

The entire crossing mass cancels.

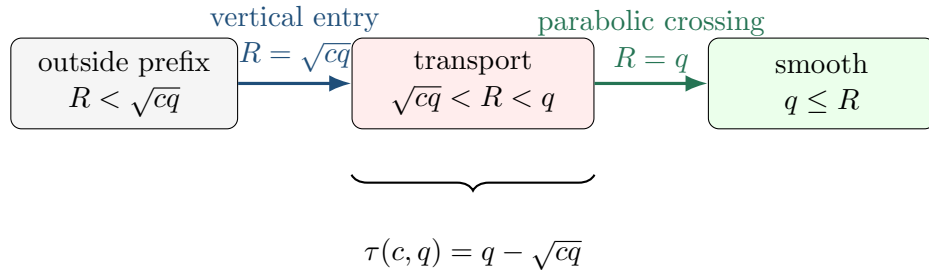


Figure 3: The exact state path of a positive-height canonical point. The second transition only reallocates signed mass between $-T$ and A , so it cancels from the residual increment.

This is the telescopic nature of the squared coordinate: the point's imaginary height determines when the moving parabola reaches it, while the crossing itself contributes zero to the increment of the total prefix mass.

10 Why $2ab$ measures both height and lifetime

Why this section is here. The same imaginary coordinate that measures factor imbalance also measures how long a source remains in transport. This is the key space–time link in the construction.

Define the signed continuous delay

$$\tau(c, q) := q - \sqrt{cq}. \quad (91)$$

It is positive exactly for transport-capable points, zero at $c = q$, and negative for points that enter already smooth.

Put

$$r := \frac{c}{q}. \quad (92)$$

Then

$$Y = \frac{q^2}{2}(1 - r^2) \quad (93)$$

and

$$\tau = q(1 - \sqrt{r}). \quad (94)$$

Since

$$1 - r^2 = (1 - \sqrt{r})(1 + \sqrt{r})(1 + r), \quad (95)$$

we obtain an exact identity.

Theorem 10.1 (Exact height–lifetime identity). *For every positive pair (c, q) ,*

$$Y = \frac{q\tau}{2} \left(1 + \sqrt{\frac{c}{q}}\right) \left(1 + \frac{c}{q}\right). \quad (96)$$

Equivalently,

$$\tau = \frac{2Y}{q \left(1 + \sqrt{c/q}\right) (1 + c/q)}. \quad (97)$$

The sign of Y is therefore exactly the sign of the continuous delay. On the transport side $0 < c/q < 1$, the denominator in Equation (97) lies between q and $4q$, giving

$$\frac{Y}{2q} \leq \tau \leq \frac{2Y}{q}. \quad (98)$$

Since

$$L(m) = q - 1 - \lfloor \sqrt{m} \rfloor \leq q - \sqrt{m} = \tau(c, q) \quad (99)$$

for transport points,

$$L(m) \leq \frac{2Y_m}{q_m}. \quad (100)$$

In squared-space coordinates, $q = \sqrt{|w| + \operatorname{Im} w}$, so

$$L(w) \leq \frac{2(\operatorname{Im} w)_+}{\sqrt{|w| + \operatorname{Im} w}}. \quad (101)$$

A point close to the real axis cannot remain in transport for many prefix steps.

11 Low-height clustering in square blocks

Why this section is here. Low height is the part of the problem that geometry alone can control. Here bounded factor gap becomes bounded occupancy, which yields a genuine theorem before any high-sector cancellation is assumed.

The n th square block becomes the vertical strip

$$\mathcal{B}_n := \{w_m : n^2 \leq \operatorname{Re} w_m < (n+1)^2\}. \quad (102)$$

For a canonical pair (c, q) , define the absolute factor gap

$$d := |q - c|. \quad (103)$$

Then

$$|Y| = \frac{|q^2 - c^2|}{2} = \frac{d(q + c)}{2}. \quad (104)$$

For a squarefree source $m > 1$, one has $d \geq 1$: equality $d = 0$ would force $m = c^2$.

Lemma 11.1 (At most one product per positive absolute gap). *Fix $n \geq 1$ and an integer $d \geq 1$. There is at most one unordered positive factor pair $\{c, q\}$ with $|q - c| = d$ and*

$$n^2 \leq cq < (n + 1)^2. \quad (105)$$

Proof. Write the smaller factor as r and the larger as $r + d$. The product is $f_d(r) = r(r + d)$. If $f_d(r) \in B_n$, then

$$2r + d \geq 2\sqrt{r(r + d)} \geq 2n.$$

Hence

$$f_d(r + 1) - f_d(r) = 2r + d + 1 \geq 2n + 1,$$

which exceeds the integer diameter $2n$ of the block. Thus two distinct values of r cannot both occur. $\square$

Theorem 11.2 (Sharpened absolute low-height clustering). *For every $H \geq 0$,*

$$\boxed{\#\{w_m \in \mathcal{B}_n : |Y_m| \leq H\} \leq \left\lfloor \frac{H}{n} \right\rfloor}. \quad (106)$$

Moreover,

$$\boxed{\#\{w_m \in \mathcal{B}_n : |Y_m| \leq n\} = 0}. \quad (107)$$

Proof. If $cq \in B_n$, then

$$c + q \geq 2\sqrt{cq} \geq 2n.$$

Therefore

$$|Y| = \frac{d(c + q)}{2} \geq dn.$$

The condition $|Y| \leq H$ implies $d \leq \lfloor H/n \rfloor$. Since $d \geq 1$, there are at most $\lfloor H/n \rfloor$ possible gaps, and [Lemma 11.1](#) gives at most one point for each gap.

If $|Y| \leq n$, the inequalities force $d = 1$ and $c + q = 2n$. But $|q - c| = 1$ makes $c + q$ odd, whereas $2n$ is even. Hence equality is impossible and the set is empty. $\square$

11.1 Parity-refined bounds

The factor gap has a prescribed parity:

- if m is odd, then c and q are odd, so d is even;
- if m is even and squarefree, then the canonical factors have opposite parity, so d is odd.

Let

$$D := \left\lfloor \frac{H}{n} \right\rfloor. \quad (108)$$

The permitted positive gaps are therefore

$$2, 4, \dots, 2 \left\lfloor \frac{D}{2} \right\rfloor$$

for odd sources and

$$1, 3, \dots, 2 \left\lceil \frac{D}{2} \right\rceil - 1$$

for even sources. Consequently,

$$\#\{w_m \in \mathcal{B}_n : m \text{ odd}, |Y_m| \leq H\} \leq \left\lfloor \frac{H}{2n} \right\rfloor, \quad (109)$$

$$\#\{w_m \in \mathcal{B}_n : m \text{ even}, |Y_m| \leq H\} \leq \left\lceil \frac{1}{2} \left\lfloor \frac{H}{n} \right\rfloor \right\rceil. \quad (110)$$

At the endpoint $H = n$, both populations are empty by [Equation \(107\)](#).

At height $H = \Lambda n$, the scale-free bounds become

$$\#\{w_m \in \mathcal{B}_n : |Y_m| \leq \Lambda n\} \leq \lfloor \Lambda \rfloor, \quad (111)$$

$$\#\{m \text{ odd} : |Y_m| \leq \Lambda n\} \leq \left\lfloor \frac{\Lambda}{2} \right\rfloor. \quad (112)$$

11.2 A sharper low-height space–time budget

For negative-height points, the transport lifetime is zero. For positive-height points in $\mathcal{B}_n$, one has $q \geq \sqrt{m} \geq n$, so [Equation \(100\)](#) gives

$$L(m) \leq \frac{2H}{n} \quad (0 < Y_m \leq H). \quad (113)$$

Therefore

$$\boxed{\sum_{\substack{w_m \in \mathcal{B}_n \\ |Y_m| \leq H}} (1 + L(m)) \leq \left\lfloor \frac{H}{n} \right\rfloor \left(1 + \frac{2H}{n}\right)}. \quad (114)$$

The parity-refined versions are

$$\sum_{\substack{w_m \in \mathcal{B}_n \\ m \text{ odd}, |Y_m| \leq H}} (1 + L(m)) \leq \left\lfloor \frac{H}{2n} \right\rfloor \left(1 + \frac{2H}{n}\right), \quad (115)$$

$$\sum_{\substack{w_m \in \mathcal{B}_n \\ m \text{ even}, |Y_m| \leq H}} (1 + L(m)) \leq \left\lceil \frac{1}{2} \left\lfloor \frac{H}{n} \right\rfloor \right\rceil \left(1 + \frac{2H}{n}\right). \quad (116)$$

At $H = \Lambda n$, all three right sides are bounded independently of n .

12 Conformal expansion and the missing two powers

Why this section is here. The square map explains the scale but not the sign. Its Jacobian supplies the required two powers of dilution; the later Gram analysis must still explain cancellation.

Consider the continuous map

$$\Phi(c, q) := (x(c, q), y(c, q)) = \left(cq, \frac{q^2 - c^2}{2} \right). \quad (117)$$

Its differential is

$$D\Phi(c, q) = \begin{pmatrix} q & c \\ -c & q \end{pmatrix}. \quad (118)$$

The two columns are orthogonal and have equal length $\sqrt{c^2 + q^2}$. Consequently,

$$D\Phi(c, q)^T D\Phi(c, q) = (c^2 + q^2)I. \quad (119)$$

Thus Φ is conformal: locally it rotates and expands every direction by the same factor

$$\lambda(c, q) = \sqrt{c^2 + q^2}. \quad (120)$$

The area Jacobian is

$$J(c, q) = \det D\Phi(c, q) = c^2 + q^2 = 2|w|. \quad (121)$$

By AM–GM,

$$c^2 + q^2 \geq 2cq = 2x. \quad (122)$$

Therefore, in the square strip $x \asymp n^2$,

$$\lambda(c, q) \gtrsim n, \quad J(c, q) \gtrsim n^2. \quad (123)$$

This is the two-dimensional version of the vertical spacing estimate. Unit-scale separation in the factor plane becomes scale- n separation in the squared plane, while unit factor-lattice area becomes scale- n^2 area. The coherent square-prefix ceiling loses exactly two powers if the relevant kernel neighborhoods have bounded squared-space area after localization.

Caution 12.1 (What the Jacobian does not prove). The Möbius point cloud is discrete and arithmetically restricted, and the square-prefix kernel is not Euclidean convolution. The Jacobian therefore does not by itself prove a signed variance estimate. It proves the exact geometric dilution scale that a valid operator argument must inherit.

12.1 Two-dimensional area versus one-dimensional cofactor curves

The conformal Jacobian governs two-dimensional regions in the continuous (c, q) factor plane. The arithmetic measure is not two-dimensional: for each fixed cofactor c , it is supported on the one-dimensional prime-sampled curve

$$q \mapsto \Phi(c, q) = \left(cq, \frac{q^2 - c^2}{2} \right), \quad q \text{ prime}, \quad q > P^+(c).$$

Along that curve,

$$\left\| \frac{\partial \Phi}{\partial q}(c, q) \right\| = \sqrt{c^2 + q^2}. \quad (124)$$

Thus one-dimensional arclength is also expanded, but the number and arithmetic placement of samples are governed by primes rather than by the area Jacobian. A two-dimensional density estimate cannot be applied directly to a single cofactor curve.

The outer curve $c = 1$ makes this obstruction explicit. Its contribution to the n th complete prefix is

$$S_n^{(1)} = -\pi(X_n), \quad (125)$$

because every prime $q \leq X_n$ has canonical pair $(1, q)$ and Möbius weight -1 . The prime number theorem therefore predicts

$$\sum_{n \leq N} |S_n^{(1)}|^2 \sim \frac{N^5}{20 \log^2 N}, \quad (126)$$

up to the usual slowly varying correction. The prime curve by itself is not on the cubic scale. Cancellation between $c = 1$ and the aggregate of the curves $c \geq 2$ is indispensable.

This also shows why a bound of the form

$$\sum_{q \leq Q} e\left(\alpha \frac{q^2 - c^2}{2}\right) \ll Q^{1/2+\varepsilon}$$

cannot hold uniformly in α : at $\alpha = 0$ the left side is $\pi(Q) + O(1)$. The zero and major-arc modes must be cancelled across cofactor curves by the Möbius weights $\mu(c)$ and the exact smooth-transport architecture. Oscillatory estimates are relevant only after that coherent component has been isolated.

12.2 Small-modulus prime resonances

The coherent rational modes are more rigid than a generic major-arc heuristic suggests.

Theorem 12.2 (Prime squares modulo 24). *If $q > 3$ is prime, then*

$$\boxed{q^2 \equiv 1 \pmod{24}}. \quad (127)$$

Proof. The prime q is odd, so $q^2 \equiv 1 \pmod{8}$. Also $3 \nmid q$, so $q \equiv \pm 1 \pmod{3}$ and $q^2 \equiv 1 \pmod{3}$. Since 8 and 3 are coprime, the Chinese remainder theorem gives Equation (127). $\square$

Corollary 12.3 (Perfect coherence on the 2–3 resonance lattice). *Let $m \in \mathbb{Z}$ and put $\alpha = m/12$. For every prime $q > 3$,*

$$e\left(\alpha \frac{q^2}{2}\right) = e\left(\frac{mq^2}{24}\right) = e\left(\frac{m}{24}\right). \quad (128)$$

Therefore, if $Q > 3$,

$$\boxed{\sum_{\substack{Q < q \leq 2Q \\ q \text{ prime}}} e\left(\alpha \frac{q^2}{2}\right) = e\left(\frac{m}{24}\right) (\pi(2Q) - \pi(Q))}. \quad (129)$$

In particular, the magnitude equals the full prime count.

This exact coherence includes

$$\alpha \in \left\{ 0, \frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{3}{4}, \frac{1}{6}, \frac{5}{6} \right\}.$$

It is not a failure of the major/minor-arc philosophy. It identifies a finite, explicit resonant component that must be reproduced by the major-arc model and then cancelled across cofactor curves.

Proposition 12.4 (The rational quadratic phase has modulus $2r$). *Let $a, r, u \in \mathbb{Z}$ with $r \neq 0$. Then*

$$\frac{e(a(u+r)^2/(2r))}{e(au^2/(2r))} = (-1)^{ar}. \quad (130)$$

Consequently $u \mapsto e(au^2/(2r))$ is always periodic modulo $2r$, but when ar is odd it is not a well-defined function of $u \bmod r$.

Proof. Expanding the square gives

$$\frac{a(u+r)^2 - au^2}{2r} = au + \frac{ar}{2}.$$

The integer term au disappears under $e(t) = e^{2\pi it}$, while $e(ar/2) = (-1)^{ar}$. □

Definition 12.5 (Corrected reduced quadratic prime factor). For $r \geq 1$ and $(a, r) = 1$, define

$$G_{2r}^\times(a) := \sum_{\substack{u \bmod 2r \\ (u, 2r)=1}} e\left(\frac{au^2}{2r}\right), \quad (131)$$

$$\gamma(a, r) := \frac{G_{2r}^\times(a)}{\varphi(2r)}. \quad (132)$$

At $a = 1, r = 3$, the units modulo 6 are 1 and 5, both with square 1 modulo 6. Thus

$$G_6^\times(1) = 2e(1/6), \quad \boxed{\gamma(1, 3) = e(1/6), \quad |\gamma(1, 3)| = 1.} \quad (133)$$

By contrast, the expression obtained by summing representatives modulo 3 is

$$e(1/6) + e(4/6) = 0.$$

That vanishing is an artifact of using a modulus on which the phase is not well-defined.

13 Vertical spacing over a fixed product

Why this section is here. The fixed-product fibre gives a direct spacing law. These estimates prevent low-height points from accumulating too densely above one square block.

Fix $m > 0$ and parameterize the full factor hyperbola by signed $b \in \mathbb{R}$:

$$a = \sqrt{m + b^2}, \quad Y_m(b) = 2b\sqrt{m + b^2}. \quad (134)$$

Then

$$Y'_m(b) = 2 \frac{m + 2b^2}{\sqrt{m + b^2}} \geq 2\sqrt{m}. \quad (135)$$

The derivative is positive for all signed b , so the map is strictly increasing from the lower to the upper half-plane.

If admissible b values differ by at least h , then

$$|\Delta Y| \geq 2h\sqrt{m}. \quad (136)$$

At square scale $m \asymp n^2$, vertical spacing is at least order n . This agrees with the conformal scale factor in [Equation \(123\)](#).

14 The proved local bound at low height

Why this section is here. This is the closed part of the bridge. Geometry alone bounds the low-height increment in each nontrivial square block, and the isolated unit source is a finite endpoint. The resulting cumulative low sector has translated-window energy of order HN^2 .

Fix $\Lambda > 0$. Use the canonical conventions

$$P^+(1) = 1, \quad c_1 = 1, \quad Y_1 = 0,$$

and assign every integer m to its square block

$$j = e(m) = \lfloor \sqrt{m} \rfloor. \quad (137)$$

Define

$$d_j^{\text{low}}(\Lambda) := \sum_{\substack{m \in B_j \\ |Y_m| \leq \Lambda j}} \mu(m), \quad (138)$$

$$d_j^{\text{high}}(\Lambda) := \sum_{\substack{m \in B_j \\ |Y_m| > \Lambda j}} \mu(m). \quad (139)$$

The predicates are complementary, so the complete block increment satisfies

$$\sum_{m \in B_j} \mu(m) = d_j^{\text{low}}(\Lambda) + d_j^{\text{high}}(\Lambda). \quad (140)$$

The block $B_0 = [0, 1)$ contains only 0, whose Möbius weight is zero, so

$$d_0^{\text{low}}(\Lambda) = d_0^{\text{high}}(\Lambda) = 0. \quad (141)$$

The unit source $m = 1$ lies in $B_1 = [1, 4)$ and is low because $Y_1 = 0$. For every block, the nontrivial low sources satisfy the occupancy bound in [Equation \(111\)](#); hence

$$\boxed{|d_j^{\text{low}}(\Lambda)| \leq \lfloor \Lambda \rfloor + \mathbf{1}_{\{j=1\}} \leq \lfloor \Lambda \rfloor + 1.} \quad (142)$$

Equivalently, one may separate the single $m = 1$ contribution and retain the sharper bound $\lfloor \Lambda \rfloor$ for the remaining nontrivial population.

Define the cumulative sectors

$$S_n^{\text{low}}(\Lambda) := \sum_{0 \leq j \leq n} d_j^{\text{low}}(\Lambda), \quad (143)$$

$$S_n^{\text{high}}(\Lambda) := \sum_{0 \leq j \leq n} d_j^{\text{high}}(\Lambda). \quad (144)$$

Summing Equation (140) over $0 \leq j \leq n$ gives the exact native decomposition

$$S_n = M((n+1)^2 - 1) = S_n^{\text{low}}(\Lambda) + S_n^{\text{high}}(\Lambda). \quad (145)$$

For $n \geq 1$, the blocks $B_1, \dots, B_n$ contain all sources up to X_n , the unit source occurs once, and therefore

$$|S_n^{\text{low}}(\Lambda)| \leq 1 + \lfloor \Lambda \rfloor n \leq (\lfloor \Lambda \rfloor + 1)n. \quad (146)$$

Consequently, if $N \geq 1$ and $1 \leq H \leq N$, then every $n \in [N, N+H)$ satisfies $n < 2N$, and hence

$$\sum_{n=N}^{N+H-1} |S_n^{\text{low}}(\Lambda)|^2 \leq 4(\lfloor \Lambda \rfloor + 1)^2 H N^2. \quad (147)$$

The global cubic consequence is

$$\sum_{n \leq N} |S_n^{\text{low}}(\Lambda)|^2 \ll_{\Lambda} N^3. \quad (148)$$

No Möbius cancellation is used.

Finally, from Equation (145),

$$|S_n|^2 \leq 2|S_n^{\text{low}}|^2 + 2|S_n^{\text{high}}|^2, \quad (149)$$

$$|S_n^{\text{high}}|^2 \leq 2|S_n|^2 + 2|S_n^{\text{low}}|^2. \quad (150)$$

These two norm inequalities, rather than an invalid energy subtraction identity, give the exact equivalence between the total and high-sector local criteria.

15 The elementary pointwise high-height bridge

Why this section is here. Everything difficult is concentrated in one finite signed sum. The translated-window, Gram, covariance, and partial-moment formulations are useful proof interfaces, but the logical burden is already complete at one point.

For fixed $\Lambda > 0$, the canonical high increment and prefix are those of Equations (139) and (144):

$$d_j^{\text{high}}(\Lambda) = \sum_{\substack{j^2 \leq m < (j+1)^2 \\ |Y_m| > \Lambda j}} \mu(m), \quad S_N^{\text{high}}(\Lambda) = \sum_{0 \leq j \leq N} d_j^{\text{high}}(\Lambda).$$

Equivalently, with $X_N = (N + 1)^2 - 1$ and $e(m) = \lfloor \sqrt{m} \rfloor$,

$$S_N^{\text{high}}(\Lambda) = \sum_{\substack{1 \leq m \leq X_N \\ |P^+(m)^2 - (m/P^+(m))^2| > 2\Lambda e(m)}} \mu(m). \quad (151)$$

Thus the remaining statement can be written without block notation.

Conjecture 15.1 (Elementary canonical pointwise estimate). *For every $\varepsilon > 0$ there is a finite constant $C_{\varepsilon, \Lambda}$ such that, for every $N \geq 1$,*

$$\left| S_N^{\text{high}}(\Lambda) \right|^2 \leq C_{\varepsilon, \Lambda} N^{2+\varepsilon}. \quad (\text{PHS})$$

Equivalently, after renaming ε ,

$$\left| S_N^{\text{high}}(\Lambda) \right| \ll_{\varepsilon, \Lambda} N^{1+\varepsilon}. \quad (152)$$

Theorem 15.2 (Elementary native bridge). *For every fixed $\Lambda > 0$, [conjecture 15.1](#) is equivalent to the Riemann Hypothesis.*

Proof. The low sector satisfies

$$|S_N^{\text{low}}(\Lambda)|^2 \ll_{\Lambda} N^2.$$

The exact recombination $S_N = S_N^{\text{low}} + S_N^{\text{high}}$ and the two inequalities

$$|S_N|^2 \leq 2|S_N^{\text{low}}|^2 + 2|S_N^{\text{high}}|^2, \quad |S_N^{\text{high}}|^2 \leq 2|S_N|^2 + 2|S_N^{\text{low}}|^2$$

show that [Equation \(PHS\)](#) is equivalent to the pointwise square-prefix Mertens estimate. By square-endpoint interpolation, that estimate is equivalent to the classical Mertens criterion and hence to RH. $\square$

15.1 Equivalent translated-window formulation

The formerly emphasized local statement is now an immediate equivalent packaging.

Conjecture 15.3 (Canonical high-sector uniform local estimate). *For every $\varepsilon > 0$ there is a finite constant $C_{\varepsilon, \Lambda}$ such that, for every $N \geq 1$ and $1 \leq H \leq N$,*

$$\sum_{n=N}^{N+H-1} \left| S_n^{\text{high}}(\Lambda) \right|^2 \leq C_{\varepsilon, \Lambda} H N^{2+\varepsilon}. \quad (\text{HS})$$

Proposition 15.4 (Pointwise-local equivalence). *For fixed $\Lambda > 0$, [conjecture 15.1](#) and [conjecture 15.3](#) are equivalent.*

Proof. The local estimate gives the pointwise estimate by taking $H = 1$. Conversely, if $N \leq n < N + H$ and $H \leq N$, then $n < 2N$. Applying [Equation \(PHS\)](#) at each n and summing gives

$$\sum_{n=N}^{N+H-1} |S_n^{\text{high}}|^2 \leq C_{\varepsilon, \Lambda} H (2N)^{2+\varepsilon},$$

and the fixed factor $2^{2+\varepsilon}$ is absorbed into the constant. $\square$

Theorem 15.5 (The local native bridge). *For every fixed $\Lambda > 0$, [conjecture 15.3](#) is equivalent to the Riemann Hypothesis.*

Proof. Combine [Proposition 15.4](#) and [Theorem 15.2](#). □

16 Why the cubic scale is natural

Why this section is here. Before asking for a proof, we check that every scale is dimensionally consistent. This section explains why N^3 , rather than the coherent N^5 ceiling, is the natural target.

The block increment

$$d_n = S_n - S_{n-1} \tag{153}$$

satisfies the trivial bound $|d_n| \leq 2n + 1$. Expanding the cumulative energy gives

$$\sum_{n \leq N} |S_n|^2 = \sum_{j, k \leq N} (N - \max\{j, k\} + 1) d_j d_k. \tag{154}$$

The fully coherent ceiling is

$$N^5 = \underbrace{N}_{\text{prefix multiplicity}} \times \underbrace{N^2}_{\text{block pairs}} \times \underbrace{N^2}_{\text{raw pair amplitude}}. \tag{155}$$

The squared map provides the exact missing scale:

$$J(c, q) = c^2 + q^2 \geq 2cq \asymp n^2. \tag{156}$$

Thus a unit area of factor combinations is diluted by n^{-2} in the squared plane. Equivalently, each coordinate direction is stretched by order n . If the localized kernel respects this conformal dilution, then the effective pair amplitude becomes order one:

$$N^3 = \underbrace{N}_{\text{prefix multiplicity}} \times \underbrace{N^2}_{\text{block pairs}} \times \underbrace{1}_{\text{squared-space interaction}}. \tag{157}$$

The low-height theorem proves this reduction directly in the region $|Y| \lesssim n$. The pointwise high-height conjecture asks for the corresponding signed cancellation in the remaining region.

17 The pointwise and local RH criteria

Why this section is here. The pointwise square-prefix estimate is the minimal RH-equivalent statement. The translated-window estimate is equivalent because it includes $H = 1$ and because pointwise bounds sum over windows with $H \leq N$.

Define

$$V_{\text{loc}}(N, H) := \sum_{n=N}^{N+H-1} |S_n|^2. \tag{158}$$

Theorem 17.1 (Elementary pointwise square-prefix criterion). *The Riemann Hypothesis is equivalent to the statement that, for every $\varepsilon > 0$,*

$$\boxed{|S_N|^2 \ll_{\varepsilon} N^{2+\varepsilon}}. \quad (159)$$

Equivalently,

$$|A_N - T_N| \ll_{\varepsilon} N^{1+\varepsilon}. \quad (160)$$

Proof. Under RH, $M(x) \ll_{\delta} x^{1/2+\delta}$ for every $\delta > 0$, and $X_N \asymp N^2$. Conversely, if $N^2 \leq x < (N+1)^2$, then

$$|M(x) - M(N^2 - 1)| \leq 2N + 1$$

because $|\mu(m)| \leq 1$. Thus the square-endpoint estimate interpolates to $M(x) \ll_{\varepsilon} x^{1/2+\varepsilon}$, the classical Mertens criterion. The $A - T$ formulation follows from [Theorem 1.1](#). $\square$

Theorem 17.2 (Uniform local square-prefix criterion). *The Riemann Hypothesis is equivalent to the statement that, for every $\varepsilon > 0$,*

$$V_{\text{loc}}(N, H) \ll_{\varepsilon} HN^{2+\varepsilon} \quad (1 \leq H \leq N). \quad (161)$$

Moreover, [Equation \(161\)](#) is equivalent directly to the pointwise estimate [Equation \(159\)](#).

Proof. The local estimate implies the pointwise estimate by setting $H = 1$. Conversely, if $N \leq n < N + H$ and $H \leq N$, then $n < 2N$. Applying [Equation \(159\)](#) to each term and summing yields

$$V_{\text{loc}}(N, H) \ll_{\varepsilon} H(2N)^{2+\varepsilon} \ll_{\varepsilon} HN^{2+\varepsilon}.$$

Now use [Theorem 17.1](#). $\square$

By [Theorem 15.2](#) and [Proposition 15.4](#), the pointwise high-sector inequality [Equation \(PHS\)](#) is the minimal native arithmetic form of the same criterion. Global cubic energy remains a natural diagnostic, but it is not needed in the logical equivalence.

18 Numerical evidence and preregistered diagnostics

Why this section is here. The computations are designed to falsify mechanisms, not to certify RH. Exact checks validate identities; asymptotic diagnostics identify where a proposed uniform theorem would first fail.

The computations reported here use exact integer arithmetic through

$$N_{\text{max}} = 4096, \quad X_{\text{max}} = (4097)^2 - 1 = 16,785,408. \quad (162)$$

A linear sieve produced

$$1,078,362 \text{ primes} \quad \text{and} \quad 10,204,280 \text{ squarefree sources } m \geq 2. \quad (163)$$

18.1 Deterministic exact checks

The implementation independently constructed S_n , A_n , and T_n from the source entry and transition intervals. Over all 4096 prefixes,

$$\max_{n \leq 4096} |S_n - (A_n - T_n)| = 0. \quad (164)$$

Across all 10,204,280 squarefree sources, the following violation counts were also zero:

Table 2: Exact squared-space verification through $X_{\max}$.

Check	Violations
$(q^2 + c^2)^2 = (2m)^2 + (q^2 - c^2)^2$	0
$c^2(q^2 - c^2) = m^2 - c^4$	0
height-lifetime inequality	0
prefix identity $S_n = A_n - T_n$	0

18.2 Small-modulus prime-phase regression test

For the toy interval $2000 < q \leq 4000$, there are exactly

$$\pi(4000) - \pi(2000) = 247 \quad (165)$$

primes. Every one satisfies $q^2 \equiv 1 \pmod{24}$. Hence, for $\alpha = m/12$,

$$S_{\text{prime}}(\alpha) := \sum_{\substack{2000 < q \leq 4000 \\ q \text{ prime}}} e\left(\alpha \frac{q^2}{2}\right) = 247e(m/24), \quad (166)$$

and therefore $|S_{\text{prime}}(\alpha)| = 247$ exactly.

Table 3: Exact prime-phase coherence on the tested 2–3 resonance lattice.

α	common phase	$ S_{\text{prime}}(\alpha) $
0	1	247
1/2	$e(1/4)$	247
1/3	$e(1/6)$	247
2/3	$e(1/3)$	247
1/4	$e(1/8)$	247
3/4	$e(3/8)$	247
1/6	$e(1/12)$	247
5/6	$e(5/12)$	247

The old modulus- r expression fails the regression test at $a = 1, r = 3$:

$$\sum_{\substack{u \bmod 3 \\ (u,3)=1}} e(u^2/6) = e(1/6) + e(4/6) = 0. \quad (167)$$

The corrected units-modulo-6 factor is

$$\frac{1}{\varphi(6)} \sum_{\substack{u \bmod 6 \\ (u,6)=1}} e(u^2/6) = e(1/6), \quad (168)$$

with unit magnitude.

The elevated value at $\alpha = 1/20$ is also arithmetic rather than a numerical artifact. For primes $q > 5$, $q^2 \equiv 1$ or $9 \pmod{40}$. In this interval the class counts are 123 and 124, respectively, so

$$|123e(1/40) + 124e(9/40)| = 199.828062 \dots \quad (169)$$

This illustrates why the complete reduced-square distribution modulo $2r$, not only the denominator r , belongs in the rational major-arc model.

18.3 Smooth–transport matching

Table 4: Exact cumulative prefix energies. Every entry is computed from all prefixes $1 \leq n \leq N$.

N	$\ A\ ^2/N^3$	$\ T\ ^2/N^3$	$\langle A, T \rangle/N^3$	$\ A - T\ ^2/N^3$	S_N
128	0.490490	0.469682	0.475665	0.008842	-18
256	1.386025	1.338898	1.356626	0.011671	4
512	3.709722	3.668905	3.685593	0.007442	38
1024	9.831997	9.807089	9.814616	0.009853	288
2048	26.661079	26.658451	26.655516	0.008497	263
4096	72.821788	73.064243	72.938028	0.009974	329

The residual remains near the cubic scale while the two component energies increase substantially.

18.4 The prime curve does not satisfy the cubic bound

Let

$$S_n^{(1)} = -\pi(X_n)$$

be the contribution of the curve $c = 1$, and put

$$S_n^{(\geq 2)} = S_n - S_n^{(1)}.$$

Then

$$\|S\|^2 = \|S^{(1)}\|^2 + \|S^{(\geq 2)}\|^2 + 2\langle S^{(1)}, S^{(\geq 2)} \rangle. \quad (170)$$

Table 5: Prime-curve versus remaining-curve energy decomposition.

N	V/N^3	$V_{c=1}/N^3$	$V_{c \geq 2}/N^3$	$2\langle c = 1, c \geq 2 \rangle/N^3$	$V/V_{c=1}$	correlation
128	0.008842	51.225450	50.965559	-102.182168	1.7260×10^{-4}	-0.999916714
256	0.011671	146.590116	146.035583	-292.614028	7.9618×10^{-5}	-0.999961911
512	0.007442	441.694929	441.211486	-882.898974	1.6848×10^{-5}	-0.999991721
1024	0.009853	1382.291036	1381.948964	-2764.230146	7.1282×10^{-6}	-0.999996443
2048	0.008497	4450.731250	4450.668331	-8901.391084	1.9091×10^{-6}	-0.999999045
4096	0.009974	14652.537064	14655.994904	-29308.521994	6.8072×10^{-7}	-0.999999667

At $N = 4096$, the prime curve has energy more than $1.46 \times 10^4 N^3$, while the total residual is approximately $9.97 \times 10^{-3} N^3$. The squared norm of the total is only 6.81×10^{-7} of the prime-curve squared norm. This is direct numerical evidence that the central phenomenon is cancellation *between* cofactor curves, not cubic behavior of the prime curve in isolation.

18.5 The dyadic cofactor cascade

For $C \geq 1$, define the partial cofactor residual

$$R_C(x) := 1 - \sum_{1 \leq c \leq C} \mu(c) \Pi_c(x), \quad (171)$$

and its square-prefix energy

$$V_C(N) := \sum_{n \leq N} |R_C(X_n)|^2. \quad (172)$$

Thus

$$R_1(x) = 1 - \pi(x),$$

and $R_C(x) = M(x)$ once C contains every canonical cofactor that can occur below x . The cascade tests how successive dyadic cofactor ranges cancel the prime curve.

Table 6: Complete dyadic cofactor cascade at $N = 4096$. The third column is $V_C(4096)/4096^3$; the fourth is V_C/V_1 . The final row is exact because the largest observed canonical cofactor is $692,835 < 2^{20}$.

C	$R_C(X_{4096})$	V_C/N^3	V_C/V_1
1	-1,078,361	14652.492160	1.000e+00
2	-513,936	3318.367550	2.265e-01
4	-127,056	198.352150	1.354e-02
8	86,580	99.704527	6.805e-03
16	-4,620	0.148513	1.014e-05
32	162,415	344.808298	2.353e-02
64	51,754	36.470907	2.489e-03
128	75,937	76.897452	5.248e-03
256	67,189	59.850472	4.085e-03
512	81,166	84.906888	5.795e-03
1,024	71,179	60.275193	4.114e-03
2,048	43,260	15.704342	1.072e-03
4,096	3,633	4.361902	2.977e-04
8,192	-39,326	14.649760	9.998e-04
16,384	-15,331	1.552681	1.060e-04
32,768	-2,326	0.051346	3.504e-06
65,536	4,622	0.142635	9.735e-06
131,072	708	0.011615	7.927e-07
262,144	206	0.009590	6.545e-07
524,288	332	0.009980	6.811e-07
1,048,576	329	0.009974	6.807e-07

The cascade is strongly nonmonotone. The first four cofactor scales reduce the normalized energy from 14652.49 at $C = 1$ to 0.1485 at $C = 16$, but the next dyadic block raises it to 344.81 at $C = 32$. Subsequent blocks repeatedly destroy and recreate coherent mass before the residual settles near the final cubic scale. This rules out a simple monotone “more cofactors means more cancellation” picture. The analytic estimate must control a signed cascade of dyadic cross-terms.

The machine-readable file `cofactor_cascade.csv` reports every dyadic C for all six prefix checkpoints $N = 128, 256, 512, 1024, 2048, 4096$.

18.6 Independent extension through $N = 2^{14}$

A separate memory-limited computation extended the exact checks and structural diagnostics to

$$N = 2^{14} = 16384, \quad X_N = (16385)^2 - 1 \approx 2.68 \times 10^8. \quad (173)$$

The larger targets $N = 2^{16}$ and 2^{17} would require sieving past approximately 4.3×10^9 and 1.7×10^{10} , respectively, and were not attempted in the available memory. No values at those scales are claimed.

The independently constructed smooth, transport, and residual values agree exactly at every tested checkpoint. At the terminal checkpoint,

$$\boxed{A_{16384} = -851896, \quad T_{16384} = -850299, \quad S_{16384} = -1597 = A_{16384} - T_{16384}.} \quad (174)$$

For a cofactor block $C < c \leq 2C$ with $C \asymp N^\sigma$, define the theorem-predicted rank-one residual

$$E_C^{\text{th}} := B_C - \lambda_\sigma a_C P, \quad \lambda_\sigma = \frac{2}{2 - \sigma}. \quad (175)$$

This is distinct from the orthogonal least-squares residual $B_C - \hat{\beta}_C P$. The following table reports the former.

Table 7: Theorem-predicted rank-one residuals at $N = 16384$. Residual energy is four to five orders of magnitude below the raw block energy throughout the tested range.

cofactor block	a_C	$\ B_C\ ^2/N^3$	$\ E_C^{\text{th}}\ ^2/N^3$	residual/raw
(1, 2]	-0.50000	45202.3	0.5166	1.1×10^{-5}
(2, 4]	-0.33333	21074.3	2.4454	1.2×10^{-4}
(4, 8]	-0.17619	6344.8	1.0401	1.6×10^{-4}
(8, 16]	+0.07026	1133.5	0.0481	4.2×10^{-5}
(16, 32]	-0.12347	3752.6	0.1394	3.7×10^{-5}
(32, 64]	+0.07872	1626.2	0.4517	2.8×10^{-4}
(64, 128]	-0.01655	77.4	0.0547	7.1×10^{-4}
(128, 256]	+0.00512	8.9	0.0052	5.8×10^{-4}
(256, 512]	-0.00911	29.4	0.0058	2.0×10^{-4}

The height-shell decomposition is more decisive. With shells defined by the ratio $|Y_m|/j$, where j is the entry block, the diagonal shell energies are:

Table 8: Height-shell diagonal energies at $N = 16384$. The recombined energy is not the sum of these rows; it includes their large signed cross terms.

shell for $ Y_m /j$	$\sum_n S_n^{[\ell]} ^2/N^3$
[0, 5)	0.00057
[5, 20)	0.0147
[20, 80)	0.259
[80, 320)	3.81
[320, 1280)	44.7
[1280, ∞)	86.3
signed recombination	0.0104

Using the rounded diagonal values in Table 8,

$$\sum_{\ell} G_{\ell\ell}(N)/N^3 \approx 135.084, \quad 2 \operatorname{Re} \sum_{\ell < m} G_{\ell m}(N)/N^3 \approx -135.074. \quad (176)$$

Thus the high sector is not small by shell. Its observed smallness is produced by nearly complete cancellation of large off-diagonal shell interactions. Any proof that takes absolute values or sums separate shell norms before recombination destroys the principal mechanism visible in the data.

Exact and secondary reduced-square modes

The exact congruence $q^2 \equiv 1 \pmod{24}$ was verified on all 4459 primes in the tested prime sample. Modulo 40, the same sample occupied exactly the two classes

$$q^2 \equiv 1 \text{ or } 9 \pmod{40}, \quad (177)$$

with counts 2224 and 2235. This is forced for primes $q > 5$ by $q^2 \equiv 1 \pmod{8}$ and $q^2 \equiv 1 \text{ or } 4 \pmod{5}$.

Proposition 18.1 (Two reduced-square classes modulo 40). *For every prime $q > 5$,*

$$\boxed{q^2 \equiv 1 \text{ or } 9 \pmod{40}}. \quad (178)$$

Accordingly, a conductor dividing 40 produces a two-mode prime phase rather than a featureless minor-arc sum. In the tested sample,

frequency	observed prime-sum magnitude
1/20	3607
3/20	1378
1/40	approximately 8–14
9/40	approximately 8–14

The same two exact phase modes can therefore interfere constructively or destructively. Here “resonant” means low-rank exact phase support, not necessarily large amplitude.

Cumulative Möbius-weighted cofactor phase magnitudes through the six blocks ending at $c = 64$ were

θ	(1, 2]	(2, 4]	(4, 8]	(8, 16]	(16, 32]	(32, 64]
1/12	1.0	1.6	3.4	4.4	8.0	13.9
1/20	1.0	1.8	2.5	5.2	7.3	10.7
1/40	1.0	2.0	1.8	2.2	0.38	2.8

(179)

The 1/40 row contains a substantial cancellation event at the block $16 < c \leq 32$, emphasizing again that the resonant main term is a signed cofactor cascade rather than a monotone positive contribution.

18.7 Low-height clustering and lifetime

For fixed Λ , define

$$C_n(\Lambda) := \#\{w_m \in \mathcal{B}_n : |Y_m| \leq \Lambda n\}, \quad (180)$$

$$I_n(\Lambda) := \sum_{\substack{w_m \in \mathcal{B}_n \\ |Y_m| \leq \Lambda n}} (1 + L(m)). \quad (181)$$

The sharpened exact bounds are

$$C_n(\Lambda) \leq \lfloor \Lambda \rfloor, \quad (182)$$

Table 9: Observed low-height maxima over $128 \leq n \leq 4096$, with the sharpened general and parity bounds.

Λ	count bound	total max	odd bound	odd max	even bound	even max	incidence bound	incidence max
1	0	0	0	0	0	0	0	0
2	2	1	1	0	1	1	10	1
4	4	2	2	1	2	1	36	4
8	8	4	4	2	4	2	136	10
16	16	6	8	3	8	3	528	26

$$I_n(\Lambda) \leq \lfloor \Lambda \rfloor (1 + 2\Lambda), \quad (183)$$

with $C_n(1) = I_n(1) = 0$. The odd and even source counts are bounded by $\lfloor \Lambda/2 \rfloor$ and $\lceil \lfloor \Lambda \rfloor / 2 \rceil$, respectively.

The zero at $\Lambda = 1$ is now explained exactly by Equation (107); it is not merely an empirical absence. The observed total and parity occupancies remain substantially below the sharpened deterministic envelopes.

18.8 Uniform local ratios

For a dyadic base N , define

$$\mathfrak{R}(N) := \max_{\substack{N \leq M \leq 2N-H \\ H \in \{1,2,4,\dots,N\}}} \frac{\sum_{n=M}^{M+H-1} |S_n|^2}{HM^2}. \quad (184)$$

Table 10: Worst local ratios from the exact range $n \leq 4096$.

base N	$\mathfrak{R}(N)$	$\mathfrak{R}(N)/N^{0.10}$	maximizing start	maximizing H
128	0.176193	0.108459	243	1
256	0.145830	0.083757	309	1
512	0.183003	0.098069	547	1
1024	0.157837	0.078918	1032	1
2048	0.166479	0.077665	2544	1

The fitted log–log slope of $\mathfrak{R}(N)$ over these five dyadic scales is

$$\hat{\theta} = -0.00494746. \quad (185)$$

This range passes the preliminary scale-stability test $\hat{\theta} \leq 0.10$, but it is not long enough to determine the asymptotic law.

19 Machine-checked formalization

Why this section is here. Formalization is used as a mathematical audit. It forces every index, cutoff, modulus, projection, sign, and quantifier to be explicit.

The accompanying Lean 4 project is available at

https://github.com/OVVO-Financial/RH_Lean.

It is pinned to mathlib v4.24.0 and imports fifty-eight theorem modules, covering the minimal bridge described below together with additional exploratory modules not needed for it. The source audit rejects `sorry`, `admit`, newly introduced axioms, opaque theorem substitutes, weakened statements, and circular RH assumptions; the full library builds with warnings treated as errors.

19.1 Arithmetic, phase, and geometry

The compiled arithmetic layer includes Möbius doubling, exact prime-square congruences, and the normalized cofactor expansion. The phase layer uses the canonical modulus $2r$, proves the shift-by- r sign law, and defines the reduced Gauss factor over $(\mathbb{Z}/2r\mathbb{Z})^\times$.

The geometry layer formalizes Fermat midpoint/half-gap coordinates, squared-complex recovery, cofactor parabolas, the $2ab$ displacement laws, conformality, and positive-branch injectivity.

19.2 Concrete square-prefix and Mertens bridge

Lean defines

$$M(x) = \sum_{m \leq x} \mu(m), \quad X_n = (n+1)^2 - 1, \quad S_n = M(X_n),$$

proves square-block interpolation, and proves the equivalence of the uniform local, pointwise square-prefix, square-prefix energy, and Mertens energy criteria. The final integration theorem accepts the classical proposition

$$\text{MertensEnergyBoundedStatement} \iff \text{RiemannHypothesisStatement}$$

directly as an ordinary theorem argument.

19.3 Native canonical high-sector module

The module

`RHLean.Proof.CanonicalHighSectorBridge`

defines the largest-prime canonical factor $q_m = P^+(m)$, the cofactor $c_m = m/q_m$, the doubled signed height $q_m^2 - c_m^2$, the exact square blocks, and the native low/high increments and prefixes. It proves

$$\text{squarePrefixMertens } n = \text{canonicalLowPrefix } \wedge n + \text{canonicalHighPrefix } \wedge n.$$

The module

`RHLean.Analysis.CanonicalLowOccupancy`

proves the elementary low-height clustering estimate unconditionally: on the nonzero-Möbius, $m > 1$ support of a fixed square block, the map from a source to its absolute factor gap $|q_m - c_m|$ is injective into $\{1, \dots, \lfloor \Lambda \rfloor\}$, using only $|q_m - c_m| \geq 1$, the exact product identity $q_m c_m = m$, and the

elementary growth bound $q_m + c_m \geq 2j$ on block j . Restoring the single source $m = 1$ gives the uniform bound $\lfloor \Lambda \rfloor + 1$, and this

`CanonicalLowIncrementControl` Λ

instance is constructed directly, with no remaining hypothesis. It defines the unresolved proposition

`CanonicalHighUniformLocalBoundedStatement` Λ .

Using the realized low-control object, Lean proves

`CanonicalHighUniformLocalBoundedStatement` Λ
 $\iff$ `SquarePrefixUniformLocalBoundedStatement`.

Given the classical Mertens–RH criterion, it then proves

`CanonicalHighUniformLocalBoundedStatement` Λ
 $\iff$ `RiemannHypothesisStatement`,

with the classical criterion as the only remaining theorem argument. This composition is axiom-free.

The remaining formal obligation is precisely:

1. prove the native pointwise high-sector estimate (PHS), equivalently the existing local statement (HS); the classical Mertens energy criterion equivalent to RH is imported as an ordinary theorem argument, not proved from scratch.

No unconditional proof of RH is asserted by the Lean development.

20 What is proved, formalized, and still open

Why this section is here. There are three different claims to track: a statement may be proved in the manuscript, formalized in Lean, or still required analytically. The canonical bridge now makes that boundary especially sharp.

The exact logical spine is therefore

$$\begin{aligned} \text{canonical definitions} &\implies \text{exact low/high recombination} \\ &\implies (\text{canonical (PHS)} \iff \text{total pointwise criterion}) \\ &\iff (\text{canonical (HS)} \iff \text{total local criterion}) \\ &\iff \text{RH}. \end{aligned}$$

The elementary low-control theorem and the classical Mertens–RH equivalence remain visible as their own obligations. The local and pointwise formulations carry the same information; the pointwise form is the more elementary statement.

The open analytic content is not a collection of unrelated conjectures. It is one pointwise statement: the complete native canonical high-prefix signal must satisfy $|S_N^{\text{high}}(\Lambda)|^2 = O_{\varepsilon, \Lambda}(N^{2+\varepsilon})$. The translated-window estimate follows by summation and returns the pointwise statement at $H = 1$. No reformulation of that quantity enlarges or narrows the minimal bridge.

Table 11: Mathematical and formal status of the program.

Layer	Mathematical status	Lean status
Square-root smooth-transport decomposition and channel telescoping	Proved	Core identities formalized
Squared-complex recovery, canonical cofactor curves, $2ab$ displacement, conformality, and square fibres	Proved	Formalized
Canonical low-height clustering and lifetime for $m > 1$	Proved	Formalized; <code>CanonicalLowIncrementControl</code> constructed with no remaining hypothesis
Isolated source $m = 1$	Finite endpoint, handled explicitly	Folded into the initial low block by convention
Exact canonical low/high square-prefix recombination	Proved	Formalized in the canonical bridge module
Pointwise total criterion $\iff$ pointwise canonical high criterion	Proved by two norm inequalities and the low bound	Exposed through the compiled square-prefix and canonical bridges
Pointwise criterion $\iff$ uniform local criterion	Proved by $H = 1$ and summation over $n < 2N$	Formalized for the concrete square-prefix sequence
Uniform local square-prefix criterion $\iff$ Mertens energy criterion	Proved in the project framework	Formalized
Classical Mertens criterion $\iff$ RH	Standard analytic theorem	Exact proposition exposed as a typed premise
Elementary pointwise high-sector estimate (PHS)	Open; equivalent to RH	Pointwise burden exposed through the compiled bridges
Translated-window high-sector estimate (HS)	Open; exactly equivalent to (PHS)	Exact proposition exposed, not assumed
Unconditional proof of RH	Not claimed	Not claimed

21 Conclusion and the remaining theorem

The main contribution of this paper is an exact reorganization of the square-prefix Möbius problem. The square-root cutoff gives

$$S_N = A_N - T_N.$$

The unique largest-prime factorization

$$q_m = P^+(m), \quad c_m = m/q_m$$

produces the squared point

$$w_m = m + i \frac{q_m^2 - c_m^2}{2},$$

which records the source integer and its canonical factor imbalance simultaneously. Fixed cofactors lie on parabolas, the prefix and smoothness conditions become moving boundaries, and their intersection is the exact transition $q = R$. Internal transport-to-smooth movement cancels identically in the arithmetic bookkeeping.

The ordinate $Y_m = (q_m^2 - c_m^2)/2$ controls factor gap, local occupancy, and transport lifetime. This closes the low sector pointwise:

$$|S_N^{\text{low}}(\Lambda)| \ll_{\Lambda} N.$$

After that elementary step, the entire RH problem in the native geometry is the single finite-sum inequality

$$\left| \sum_{\substack{1 \leq m \leq (N+1)^2 - 1 \\ |P^+(m)^2 - (m/P^+(m))^2| > 2\Lambda \lfloor \sqrt{m} \rfloor}} \mu(m) \right|^2 \ll_{\varepsilon, \Lambda} N^{2+\varepsilon}. \quad (\text{PHS})$$

No window, covariance, shell, probability space, or operator is needed to state it. Equivalently, before removing the elementary low sector, the total smooth-transport target is

$$|A_N - T_N| \ll_{\varepsilon} N^{1+\varepsilon}. \quad (186)$$

The translated-window estimate

$$\sum_{n=N}^{N+H-1} |S_n^{\text{high}}(\Lambda)|^2 \ll_{\varepsilon, \Lambda} H N^{2+\varepsilon}$$

is exactly equivalent: choose $H = 1$ in one direction and sum the pointwise estimate over $n < 2N$ in the other.

The accompanying Lean development machine-checks the native canonical definitions, the elementary low-height occupancy bound and the resulting unconditional low-increment control, the exact recombination with the square-prefix Mertens sequence, the total/high local equivalence, and the final conditional composition with RH. No project axiom is introduced. The remaining work is correspondingly precise:

1. prove the elementary pointwise high-sector inequality (PHS);

2. supply the classical Mertens–RH theorem in Lean.

Thus the paper does not claim an unconditional proof of RH. It gives an exact arithmetic and geometric reduction and identifies the most elementary remaining inequality: one square-prefix, one canonical high-sector selector, and one square-root-scale bound.

A Notation

Symbol	Meaning
B_n	square block $[n^2, (n+1)^2)$
R_n	square-root cutoff $n+1$
X_n	complete-prefix endpoint $R_n^2 - 1$
S_n	$M(X_n)$
A_n	signed R_n -smooth mass
T_n	one-large-prime cofactor transport statistic
c	canonical cofactor $m/P^+(m)$
q	canonical largest prime $P^+(m)$
a, b	signed Fermat coordinates $(c+q)/2, (q-c)/2$
$w = x + iy$	squared point $cq + i(q^2 - c^2)/2$
$Y = y$	signed ordinate $2ab$
ρ	$ w = (c^2 + q^2)/2$
U, V	null coordinates c^2, q^2
γ_c	fixed cofactor parabola
Γ_R	moving smooth–transport cutoff parabola
$e(m)$	first complete-prefix index containing m
$h(m)$	smoothness-transition index $q-1$
$L(m)$	discrete transport lifetime
$\tau(c, q)$	signed continuous delay $q - \sqrt{cq}$
K_N	cumulative square-prefix kernel

B Reproducibility

The numerical tables in [Section 18](#) are reproduced by the standalone script

```
python squared_space_reproducibility_v3.py \
  --nmax 4096 \
  --out squared_space_repro_v2
```

using Python, NumPy, and Numba. No network data, probabilistic sampling, or floating-point primality test is used.

B.1 Deterministic algorithm

1. Set

$$X_{\max} = (N_{\max} + 1)^2 - 1.$$

2. Run a linear sieve through $X_{\max}$ to compute:

$$\mu(m), \quad \text{spf}(m), \quad P^+(m), \quad \pi(m).$$

3. Form

$$S_n = M((n + 1)^2 - 1)$$

by integer prefix summation.

4. Construct A_n and T_n independently from the source entry/transition intervals:

$$e(m) = \lfloor \sqrt{m} \rfloor, \quad h(m) = P^+(m) - 1.$$

A squarefree source contributes to T on $[e, h)$ and to A on $[\max\{e, h\}, N_{\max}]$.

5. For every squarefree $m > 1$, put

$$q = P^+(m), \quad c = m/q, \quad 2Y = q^2 - c^2.$$

Verify the norm, curve, and lifetime identities using integers only.

6. Compute the cofactor-1 curve exactly as

$$S_n^{(1)} = -\pi(X_n),$$

then set

$$S_n^{(\geq 2)} = S_n - S_n^{(1)}$$

and evaluate all three terms in [Equation \(170\)](#).

7. Bin every squarefree source by the least dyadic threshold $C = 2^j$ containing its canonical cofactor $c = m/P^+(m)$. Cumulatively sum the bins to obtain $R_C(X_n)$ and $V_C(N)$ for every dyadic C and every reported N .

8. Evaluate the total, odd, and even low-height counts and incidences for $\Lambda = 1, 2, 4, 8, 16$.

9. Run the exact major-arc regression tests:

$$q^2 \equiv 1 \pmod{24} \quad (2000 < q \leq 4000),$$

verify the eight full-coherence frequencies in [Table 3](#), verify the old modulus-3 expression is zero, and verify the corrected normalized modulus-6 factor has unit norm.

10. Evaluate $\Re(N)$ for dyadic bases 128, 256, 512, 1024, 2048 and all dyadic window lengths $H \leq N$.

Table 12: Machine-readable reproducibility outputs.

File	Contents
run_summary.txt	range, source counts, exact violation counts, and local-ratio slope
prefix_energy.csv	S_N , total energy, and A - T component energies
prime_curve_decomposition.csv	$c = 1$, $c \geq 2$, cross term, correlation, and residual ratio
cofactor_cascade.csv	$R_C(X_N)$ and $V_C(N)$ for every dyadic C and reported N
low_height_diagnostics.csv	sharpened total/parity bounds and observed clustering/lifetime maxima
local_ratio_diagnostics.csv	worst local ratios and maximizing intervals
major_arc_regression.csv	exact mod-24 checks, rational test frequencies, and old-versus-corrected Gauss factors

B.2 Generated output files

The script writes:

For the reported run, the summary must read:

```

N_max: 4096
X_max: 16785408
prime_count: 1078362
squarefree_sources_m_ge_2: 10204280
max_observed_canonical_cofactor: 692835
max_dyadic_cofactor_threshold: 16777216
max_abs_S_minus_A_plus_T: 0
nonzero_prefix_identity_errors: 0
norm_identity_violations: 0
cofactor_curve_identity_violations: 0
height_lifetime_violations: 0
prime_square_mod24_violations_2000_4000: 0
old_modulus_3_factor_abs: 0
corrected_modulus_6_factor_abs: 1
cofactor_cascade_full_error: 0
local_ratio_log_log_slope: -0.004947461592

```

B.3 Extended $N = 16384$ diagnostic run

The extended values in [Section 18.6](#) were produced by a separate memory-limited computation reaching $X_N \approx 2.68 \times 10^8$. They are presented as numerical diagnostics, not as machine-checked theorems. A release-grade archive should include the exact source, compiler and package versions, all integer checkpoints, the shell Gram matrix, and cryptographic hashes. Until those files are attached, only the algebraic congruence statements and identities proved elsewhere in the paper have unconditional theorem status.

B.4 Integrity hashes

The SHA-256 hashes of the revised script and machine-readable outputs are listed in the release archive and in the accompanying `SHA256SUMS.txt`. This avoids copying stale hashes into the manuscript when the numerical protocol is extended.

The exact checks are binary: any nonzero violation invalidates the run. The asymptotic diagnostics are not proofs; their purpose is to identify which geometric or analytic component fails first as the range is extended.

References

- [1] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Oxford University Press, 1986.
- [2] H. L. Montgomery and R. C. Vaughan, *Multiplicative Number Theory I: Classical Theory*, Cambridge University Press, 2007.
- [3] T. Tao and V. Vu, *Additive Combinatorics*, Cambridge University Press, 2006.
- [4] F. Viole, *Complex Space Factorization*, working paper.
- [5] P. Berck and J. Hihn, Using the semivariance to estimate safety-first rules, *American Journal of Agricultural Economics* **64** (1982), 298–300.
- [6] J. A. Atwood, Demonstration of the use of lower partial moments to improve safety-first probability limits, *American Journal of Agricultural Economics* **67** (1985), 880–886.